MLFF-DATA7: Fitted Metrics, Atomic References, Objectives, and Selection

mdstats project

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MLFF-DATA7: Fitted metrics, E0 fits, objectives, and deterministic selection

Status and scope

MLFF-DATA7 is the first stage that may fit statistics to an authorized training domain and reduce that domain to a selected training subset. It consumes the immutable DATA4-DATA6 evidence catalogs and introduces no MACE training file, replay dataset, checkpoint, or test evaluation.

Every fitted object is local to exactly one canonical DATA5 training domain:

- the final development domain of one label domain; or
- the gradient-training domain of one cross-validation fold.

Outer checkpoint monitors, uncertainty-calibration cohorts, cross-validation checkpoint monitors, held-out evaluation folds, locked tests, purge units, and excluded frames SHALL NOT contribute to feature fitting, E0 fitting, difficulty ranking, or selection.

The focused implementation tests use the supplied ASE 3.29.0 source distribution. ASE remains an external runtime dependency and is not bundled.

Public contracts

DATA7 SHALL provide the following immutable records.

`FeatureFitDomain`

Binds a final-development or cross-validation-training domain to its DATA5 partition, unit IDs, frame UIDs, label domain, and optional fold index. Canonical domains are derived from DATA5 rather than supplied ad hoc.

`FeatureMetricPolicyTemplate`

Defines the static metric algorithm without fitted values:

- enabled feature blocks;
- robust or standard centering/scaling;
- missing-value indicators;
- per-block weight;
- dimension normalization;
- optional per-block principal-component cap;
- deterministic SVD sign convention and tie rules.

The initial implementation supports the blocks `raw_physical`, `lta_frame`, `mace_summary`, and `difficulty`. Model-derived blocks are optional and require the matching checkpoint-bound DATA6 artifacts.

`FittedFeatureMetric`

Stores fold-local or final-domain fitted block transforms and one transformed frame vector for every frame in the fit domain. Robust scaling uses

$$z_j = \frac{x_j - \mathrm{median}(x_j)}{\max(\mathrm{IQR}(x_j), s_{\min})}.$$

Standard scaling may instead use the mean and population standard deviation. After optional PCA, each block is multiplied by

$$\sqrt{w_b/d_b},$$

where (w_b) is the declared block weight and (d_b) the retained block dimension. This prevents a high-dimensional descriptor block from dominating solely because it has more coordinates.

PCA is fit by deterministic SVD. Each retained component is sign-normalized so its largest-magnitude loading is positive. Ties are resolved by the smallest feature index.

`AtomicReferenceFitRecord`

Fits explicit elemental reference energies only on the authorized training domain. For count matrix (A) and total-energy vector (y), the default fit is the minimum-norm least-squares solution

$$\hat{\mathbf{e}}_0 = A^+ y.$$

Optional nonnegative ridge regularization solves

$$\min_{\mathbf{e}_0} \|A\mathbf{e}_0 - y\|_2^2 + \lambda \|\mathbf{e}_0 - \mathbf{e}_{0,\mathrm{prior}}\|_2^2.$$

The record stores element order, explicit atomic-number mapping, rank, singular values, null-space dimension, residual statistics, and transfer warnings. Rank-deficient fits are allowed only when the policy explicitly permits a fixed-composition-domain minimum-norm fit. Numerical E0 values are not assigned physical meaning in the null space.

Training objective and weight contracts

`TrainingObjectivePolicy` defines energy, force, and stress loss weights and records the absence or presence of a species-aware custom force objective. `ConfigurationWeightPolicy` defines condition balancing, event emphasis, and quality modifiers. `TrainingWeightCatalog` stores per-frame configuration, energy, force, and stress weights. Weights are normalized to a mean configuration weight of one within each training domain.

Selection, exposure, and loss weight remain separate decisions.

`CheckpointMetricPolicy`

Defines the metrics and constraints later used by DATA8-DATA9 checkpoint selection. The initial record includes global target force RMSE, Li/Na/K force constraints, energy, stress, worst-condition limits, and replay-retention limits. DATA7 defines the reproducible policy but does not evaluate a trained checkpoint.

`SelectionBudgetPolicy`

Defines target training sizes and quota fractions for:

- mandatory condition anchors;
- representative anchors;
- Li/Na/K environment coverage;

- protected rare events;
- global fitted-metric FPS;
- foundation-model difficulty enrichment.

Fractions are interpreted as a deterministic weighted scheduling policy, not as independent post-hoc selections. Mandatory anchors are inserted first. The remaining category queues are interleaved by largest quota deficit. Duplicate frames are skipped without consuming a category quota.

`TrainingSelectionPlan`

Stores one deterministic master ordering with per-frame selection reasons. Each requested training set is a strict prefix:

$$\mathcal{T}_{n_1} \subset \mathcal{T}_{n_2} \subset \cdots .$$

The smallest legal target size is at least the mandatory-anchor count. An infeasible requested size fails closed rather than silently dropping a required condition.

`SelectionCoverageReport`

For every ladder level, reports:

- selected frame count;
- represented condition strata;
- represented Li/Na/K environment classes;
- protected-event coverage;
- nearest-selected fitted-metric distance quantiles;
- maximum covering radius;
- selected-to-selected redundancy quantiles.

Coverage is descriptive evidence, not a guarantee of MLFF accuracy. The ladder must later be evaluated through protocol-matched MACE learning curves.

`Data7PreparationBundle`

Binds the DATA4-DATA6 lineages, canonical training domain, fitted metric, E0 fit, objective/weight policies, checkpoint policy, selection plan, and coverage reports. One bundle is produced per final development domain and per cross-validation training fold.

Feature construction

`raw_physical` includes finite DATA4 thermodynamic, force, stress, cell, and strain scalars with explicit missing indicators. Total energy is excluded from the default distance metric because composition-dependent offsets can dominate geometry coverage; energy per atom may be enabled explicitly.

`lta_frame` uses DATA6 named frame descriptors and their missing mask.

`mace_summary` reads the checkpoint-bound per-atom descriptor sidecar and constructs deterministic configuration summaries: global mean and standard deviation plus per-Li, per-Na, and per-K means when present. Missing species are represented by zeros and explicit presence indicators.

`difficulty` uses only DATA6 training-domain residuals. It SHALL never be materialized from blinded monitor, calibration, evaluation, or locked-test records.

Selection algorithm

1. Derive one canonical `FeatureFitDomain` from DATA5.
2. Assemble raw feature blocks only for that domain.
3. Fit the block-local scalers and optional PCA using that domain only.
4. Fit the domain-local atomic reference mapping.

5. Construct condition strata from DATA5 partition conditions.
6. Add one mandatory deterministic anchor per represented stratum.
7. Build representative, species-environment, rare-event, fitted-metric FPS, and difficulty queues.
8. Interleave queues according to `SelectionBudgetPolicy` while preserving deterministic tie ordering by frame UID.
9. Fill any residual budget by global FPS, then stable frame-UID order.
10. Define every requested dataset as a prefix of the one master order.
11. Compute coverage reports without inspecting any held-out evidence.

Farthest-point sampling selects

$$i^* = \arg\max_{i \notin S} \min_{j \in S} d(\mathbf{z}_i, \mathbf{z}_j).$$

Ties within floating-point tolerance are resolved by lexicographically smallest frame UID.

Failure rules

DATA7 fails closed when:

- a requested domain is not canonical DATA5 training evidence;
- a fitted feature block is unavailable or checkpoint-incompatible;
- a fitted transform receives a monitor, calibration, evaluation, purge, or locked-test frame;
- an E0 fit lacks energy labels or elemental support;
- rank deficiency is disallowed by policy;
- a requested ladder size is smaller than mandatory coverage;
- sidecar digests or lineage records do not match;
- target sizes are not strictly increasing;
- selection cannot produce the requested largest size.

Non-goals

DATA7 does not write extended XYZ, select replay data, run MACE, choose a checkpoint, rotate epoch datasets, activate tests, or perform active learning. Those remain DATA8-DATA10 responsibilities.

DATA9A7d amendment: focus groups and generic extension coverage

DATA7 selection and decision policies no longer define cations as a universal material class. Atom groups marked with `mlff_focus`, `training_focus`, or `validation_focus` may receive explicit difficulty, environment-coverage, objective, or checkpoint treatment. If no focus group is declared, all species present in the authorized domain are treated uniformly.

`TrainingObjectivePolicy` and `CheckpointMetricPolicy` use generic focus-group and focus-atomic-number fields. Historical cation-named fields are accepted only when restoring v1 evidence and are exposed as deprecated Python aliases.

Selection coverage records generic `represented_environment_classes` supplied by optional profile providers. LTA site classes are one namespaced provider output, not a core coverage schema.

DATA9A9b production orchestration

DATA9A9b does not alter DATA7 fitting or selection mathematics. It freezes every canonical final-development and fold-training `FeatureFitDomain` plus the exact DATA7 policy set, writes one native `Data7PreparationBundle` per domain, and binds each file through a `ProductionData7ArtifactRecord`. Restart may reuse a bundle only when its file SHA-256, bundle digest, domain digest, DATA6 lineage, and native serialization all verify. Any invalid DATA7 artifact invalidates the production DATA8 tree.